ORIE 4154: Pricing and Market Design **Fall 2026**

Lecture 5 — Scarcity and the Value of Capacity: Littlewood's Rule

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5.1 Overview

So far, we have chosen a price for the customer currently in front of us. We now introduce a new source of difficulty:

Selling to one customer may prevent us from selling to a better customer later.

Airline seats, hotel-room nights, advertising impressions, and event tickets share three features:

(i) capacity is limited;

(ii) requests arrive over time and must be accepted or rejected;

(iii) unused capacity expires at the end of the horizon.

Our central economic question is therefore:

$$\boxed{\text{Why might it be optimal to reject a profitable customer today?}}$$

We will answer it exactly in a small dynamic program. The answer will take the form of an *opportunity cost*: accept a request only when its reward exceeds the marginal value of the capacity it consumes. Littlewood's rule is the canonical two-fare-class version of this principle.

The main path is

$$\boxed{\text{Littlewood's rule} \longrightarrow \text{displaced future value} \longrightarrow \Delta_t(b) \longrightarrow \text{network scarcity.}}$$

5.2 A two-stage capacity-allocation problem

A seller has C identical units of capacity. There are two customer classes:

- low-fare customers pay r_L and arrive early;
- high-fare customers pay $r_H > r_L$ and arrive later.

The later high-fare demand is the integer-valued random variable D_H. The seller knows its distribution but not its realization.

Suppose an early low-fare customer arrives when y units remain. Accepting produces r_L now, but leaves only $y - 1$ units for high-fare demand. Rejecting preserves all y units.

The expected high-fare revenue with y protected units is

$$r_H \mathbb{E}[\min\{D_H, y\}].$$

The value of protecting the yth unit is therefore

$$\begin{aligned} &r_H \mathbb{E}[\min\{D_H, y\} - \min\{D_H, y-1\}] \\ &= r_H \mathbb{P}(D_H \geq y). \end{aligned}$$

The equality follows because the yth unit is used precisely when high-fare demand reaches at least y.

Proposition 1 (Littlewood's rule). *When y units remain, accept an early low-fare request if and only if*

$$r_L \geq r_H \mathbb{P}(D_H \geq y). \tag{5.1}$$

Equivalently, protect units for high-fare demand as long as their expected marginal value exceeds the low fare.

This is already a dynamic pricing statement, even though the fares are fixed:

$$\boxed{\text{price of capacity} = \text{expected revenue displaced by using it now.}}$$

5.3 A numerical Littlewood example

Let

$$r_L = 100, \qquad r_H = 250, \qquad D_H \sim \text{Binomial}(3, 1/2).$$

The relevant tail probabilities are

y	1	2	3
$\mathbb{P}(D_H \geq y)$	7/8	1/2	1/8
$r_H \mathbb{P}(D_H \geq y)$	218.75	125	31.25.

Thus:

- with one unit remaining, reject the low fare: $218.75 > 100$;

- with two units remaining, reject the low fare: $125 > 100$;
- with three units remaining, accept the low fare: $31.25 < 100$.

The optimal protection level is two units. Equivalently,

$$y^* = \max\left\{y : \mathbb{P}(D_H \geq y) > \frac{r_L}{r_H}\right\} = 2.$$

Remark 5.1 (Why capacity matters even though y^* does not depend on C). *The number protected is determined by the high-fare demand distribution and the fare ratio. Total capacity still determines how many units are available for early customers: the corresponding low-fare booking limit is $C - y^*$.*

5.4 The dynamic-programming viewpoint

Littlewood's model separates the two customer classes in time. We now allow different types to arrive in any period.

There are t periods remaining and b units of capacity. In a period:

- a type-j request arrives with probability λ_j and offers reward r_j;
- no request arrives with probability λ_0;
- at most one unit can be sold;
- accept/reject decisions are irrevocable.

Let $V_t(b)$ be the maximum expected revenue with t periods and b units remaining. The boundary conditions are

$$V_0(b) = 0, \qquad V_t(0) = 0.$$

The Bellman equation is

$$V_t(b) = \lambda_0 V_{t-1}(b) + \sum_j \lambda_j \max\left\{V_{t-1}(b), r_j + V_{t-1}(b-1)\right\}. \tag{5.2}$$

The two terms inside the maximum are:

$$\begin{aligned} \text{reject: } & V_{t-1}(b), \\ \text{accept: } & r_j + V_{t-1}(b-1). \end{aligned}$$

Therefore type j is accepted precisely when

$$r_j \geq V_{t-1}(b) - V_{t-1}(b-1). \tag{5.3}$$

Define the marginal value of capacity

$$\Delta_{t-1}(b) = V_{t-1}(b) - V_{t-1}(b-1).$$

Then the optimal policy is simply

$$\boxed{\text{accept type } j \text{ if } r_j \geq \Delta_{t-1}(b).}$$

5.5 Executing a small dynamic program

Consider a horizon of three periods and capacity at most two. In each period:

arrival	H	L	none
probability	0.3	0.5	0.2
reward	10	4	0.

We compute the value function backward.

5.5.1 One period remaining

With positive capacity, accept either customer:

$$V_1(1) = V_1(2) = 0.3(10) + 0.5(4) = 5.$$

5.5.2 Two periods remaining

With one unit, its marginal continuation value is $V_1(1) - V_1(0) = 5$. Accept H but reject L:

$$V_2(1) = 0.3(10) + 0.5(5) + 0.2(5) = 6.5.$$

With two units, using one unit costs no future revenue because $V_1(2) - V_1(1) = 0$. Accept both types:

$$V_2(2) = 0.3(10 + 5) + 0.5(4 + 5) + 0.2(5) = 10.$$

5.5.3 Three periods remaining

With one unit, the marginal continuation value is 6.5, so reject L:

$$V_3(1) = 0.3(10) + 0.5(6.5) + 0.2(6.5) = 7.55.$$

With two units, the marginal continuation value is

$$V_2(2) - V_2(1) = 10 - 6.5 = 3.5.$$

Now the low reward 4 exceeds the opportunity cost, so accept L:

$$V_3(2) = 0.3(10 + 6.5) + 0.5(4 + 6.5) + 0.2(10) = 12.2.$$

Collecting the values:

	$b = 0$	$b = 1$	$b = 2$
$t = 0$	0	0	0
$t = 1$	0	5	5
$t = 2$	0	6.5	10
$t = 3$	0	7.55	12.2

The low-fare decision depends on both time and capacity. We reject a low request when capacity is scarce relative to the remaining opportunities, but accept it when capacity is plentiful.

5.6 What the exact DP teaches us

This is the small-system regime: the state can be enumerated, so dynamic programming gives the exact marginal value of capacity. The DP provides three ideas that survive when the system grows and the recursion itself becomes impossible to compute.

5.6.1 1. Capacity has a state-dependent price

The quantity

$$\Delta_t(b) = V_t(b) - V_t(b - 1)$$

is the opportunity cost of one unit. It is high when capacity is scarce and future opportunities are valuable.

5.6.2 2. The relevant comparison is reward versus displacement

A sale is profitable in isolation whenever $r_j > 0$. It is desirable dynamically only when

$$r_j \geq \text{future value displaced.}$$

This is why "accept every profitable customer" is generally wrong.

5.6.3 3. The DP is an exact benchmark, not always a practical algorithm

For one resource with capacity C, the DP has roughly $T(C+1)$ states. With m resources and capacity vector $(C_1, \dots, C_m)$, the number of capacity states becomes

$$\prod_{i=1}^{m}(C_i + 1).$$

This is the curse of dimensionality.

5.7 From one resource to a network

Suppose a request consumes several resources at once:

- an itinerary uses seats on multiple flight legs;
- a hotel stay uses a room on several nights;
- an advertisement consumes several campaign budgets;
- a project requires hours from several teams.

The economic logic remains unchanged. For a product consuming resource vector a_j, the exact acceptance condition is

$$r_j \geq V_{t-1}(b) - V_{t-1}(b - a_j).$$

But computing the value function is now prohibitive.

Next lecture we will replace the exact value function by a tractable optimization benchmark. This will produce two different ways to think about decisions:

1. use dual variables as approximate opportunity prices;
2. use a fluid solution as a prediction of what an offline clairvoyant would allocate.

5.8 Takeaways

1. Scarcity makes a current sale interact with future demand.
2. Littlewood's rule compares a sure low fare with the expected marginal value of protected capacity.

3. Dynamic programming makes the opportunity cost exact:

$$\text{accept reward} \geq \text{marginal continuation value.}$$

4. The value of capacity depends on both remaining time and remaining inventory.
5. Exact DP becomes computationally infeasible in resource networks, motivating fluid and offline approximations.

Suggested reading

Talluri and van Ryzin, *The Theory and Practice of Revenue Management*, Sections 2.1.1–2.2.2, for two-class capacity control and Littlewood's rule. The finite-horizon Bellman formulation here is also a compact version of the dynamic-programming development in the earlier ORIE 4154 materials.